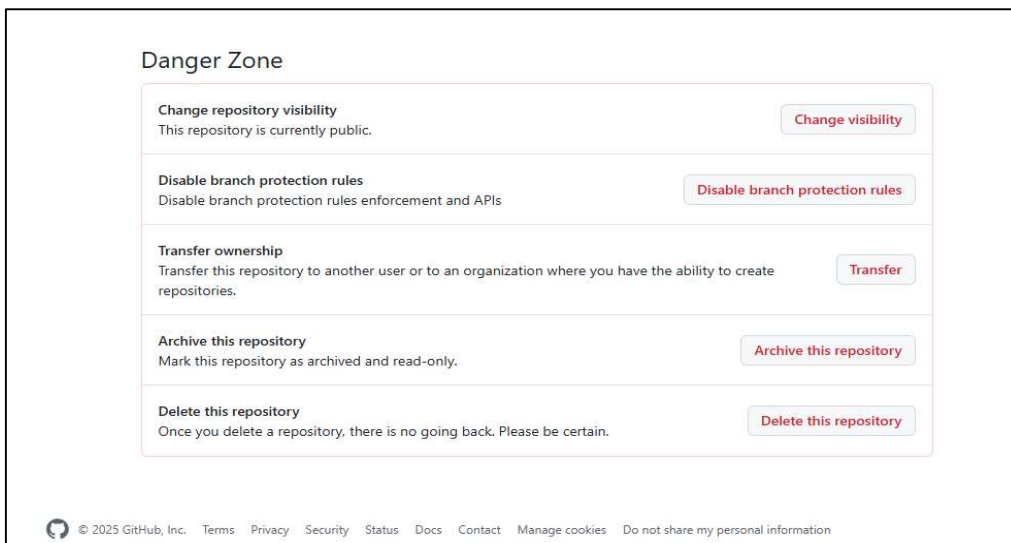
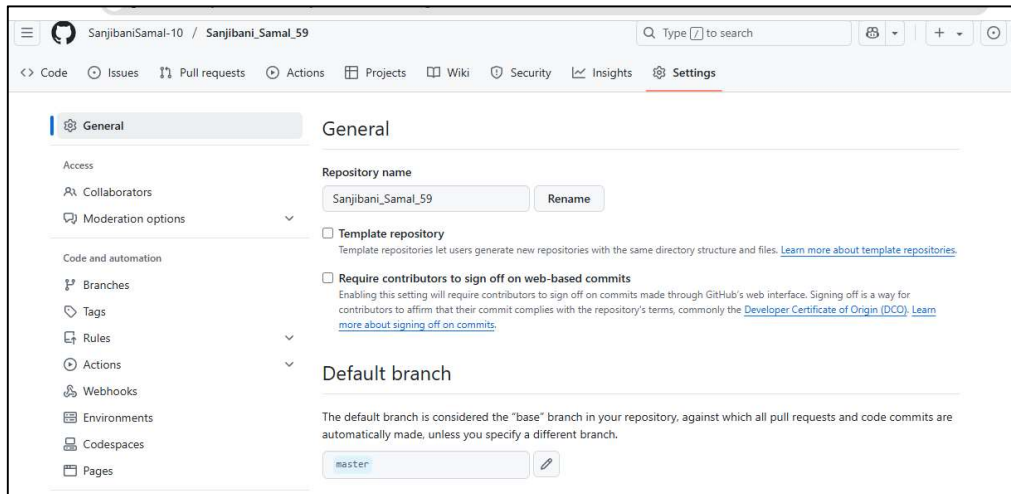


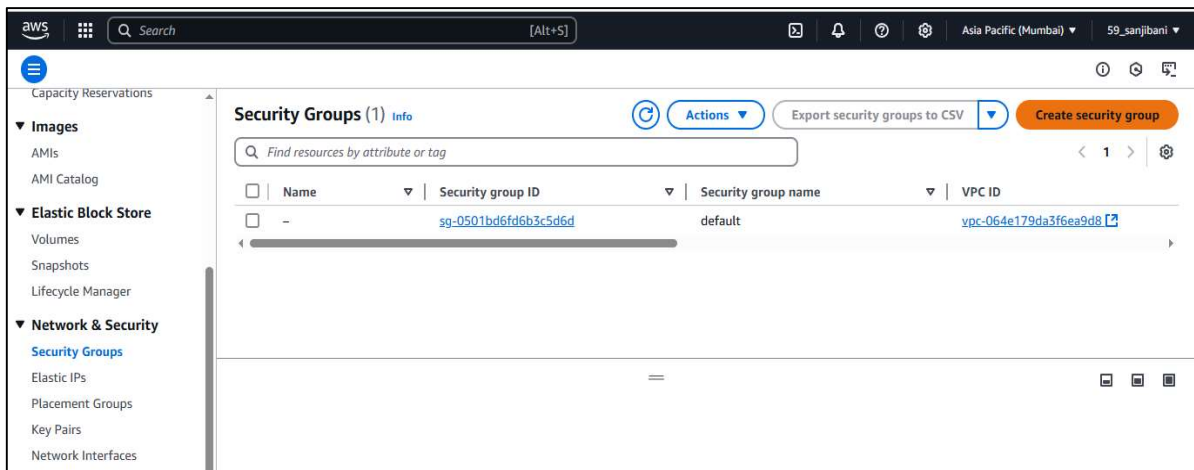
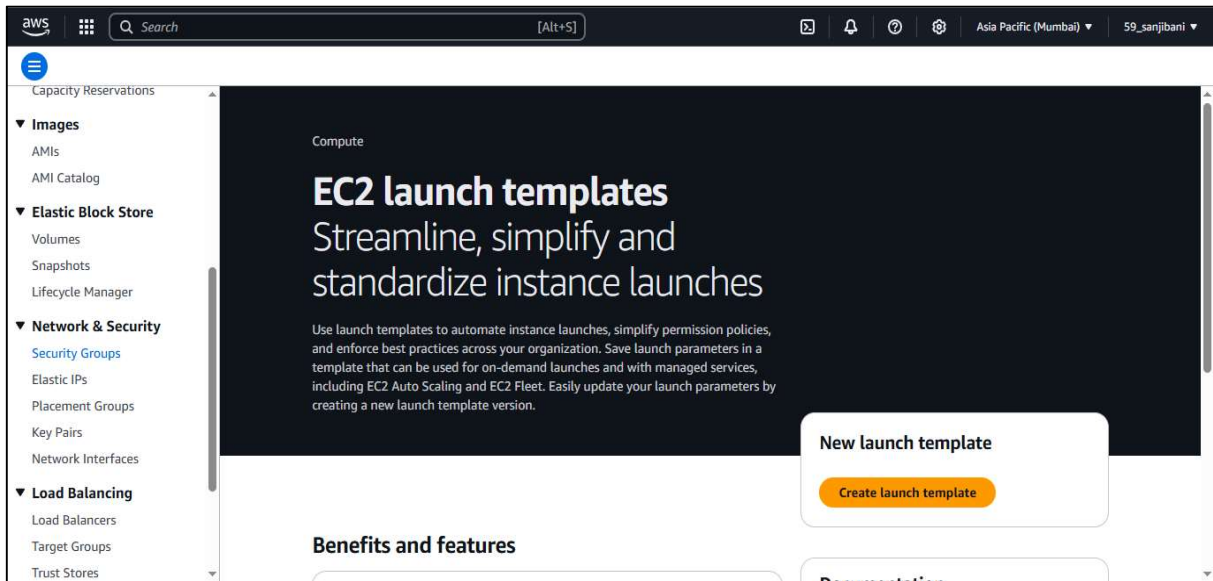
ASSIGNMENT NO :- 11

TITLE :- Build scaling plans in AWS that balance the load on different EC2 instances.

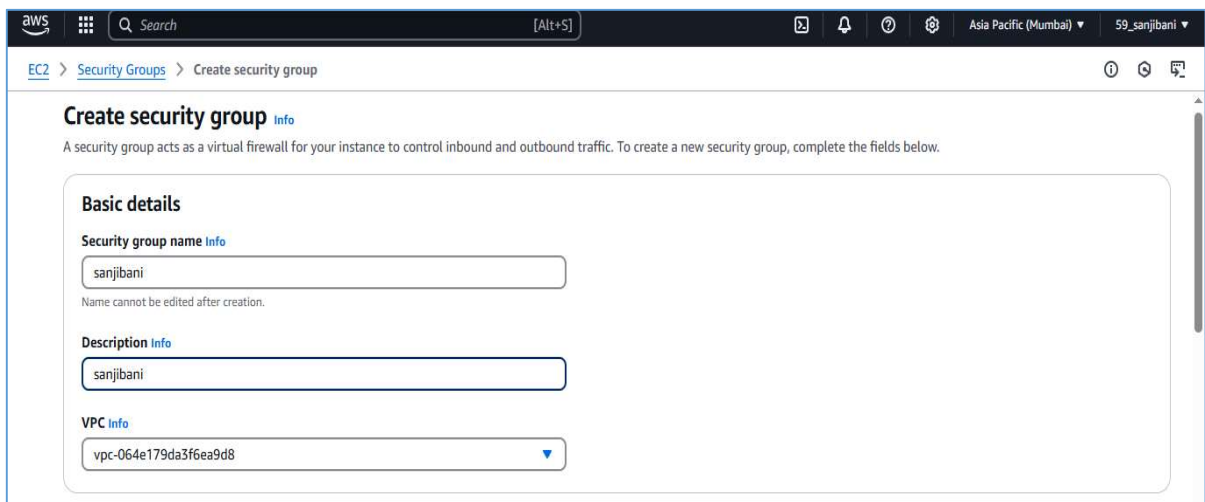
STEP 1 :- Check if your GitHub Repository is public or not . For that Login to your **GitHub Account** and then select your Repository and then go to **Setting** and navigate to **Danger Zone** and check your **Repository Visibility**.



STEP 2 :- Login to your **AWS Account** and click on **EC2** and go to **Security Groups** and click on **Create Security Group**



>> Give the **Security Group Name** and **Description** and the click on **Add Rules** of Inbound Rules



>> Add SSH , HTTP , HTTPS and CUSTOM TCP Type Rules as Given below

The screenshot shows the 'Create security group' page in the AWS Management Console. The 'Inbound rules' section is expanded, displaying a table of rules. Each rule has a 'Type' dropdown, a 'Protocol' dropdown, a 'Port range' input, a 'Source' dropdown, a 'Destination' input, and a 'Description' input. The rules are: SSH (TCP, 22), HTTP (TCP, 80), HTTPS (TCP, 443), and Custom TCP (TCP, 4000). All rules have 'Any' as the source and '0.0.0.0/0' as the destination. There are 'Delete' buttons for each rule and an 'Add rule' button at the bottom.

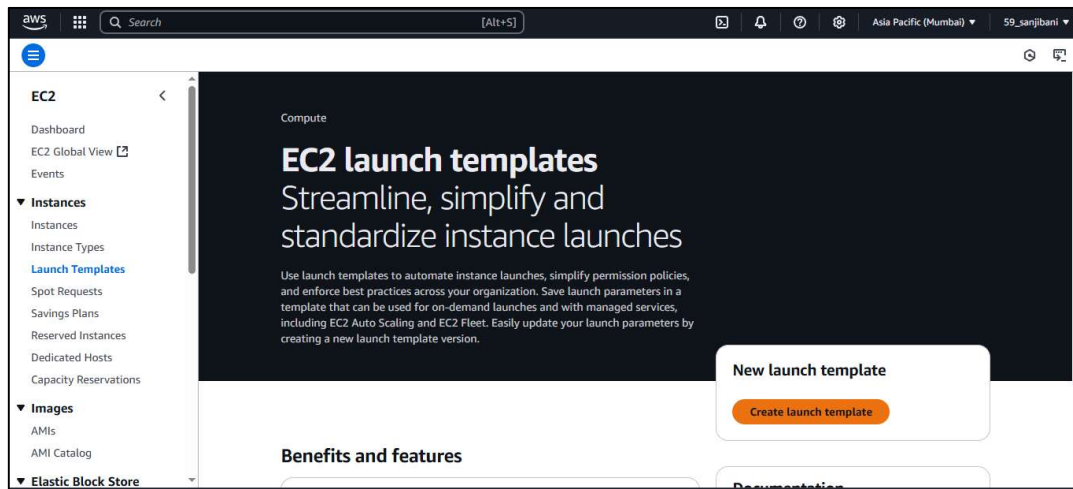
>> Click on **CREATE Security Group**

The screenshot shows the 'Create security group' page with the 'All traffic' dropdown selected. A yellow warning message is displayed: 'Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.' The 'Tags - optional' section is also visible, showing 'No tags associated with the resource.' The 'Create security group' button is highlighted in orange.

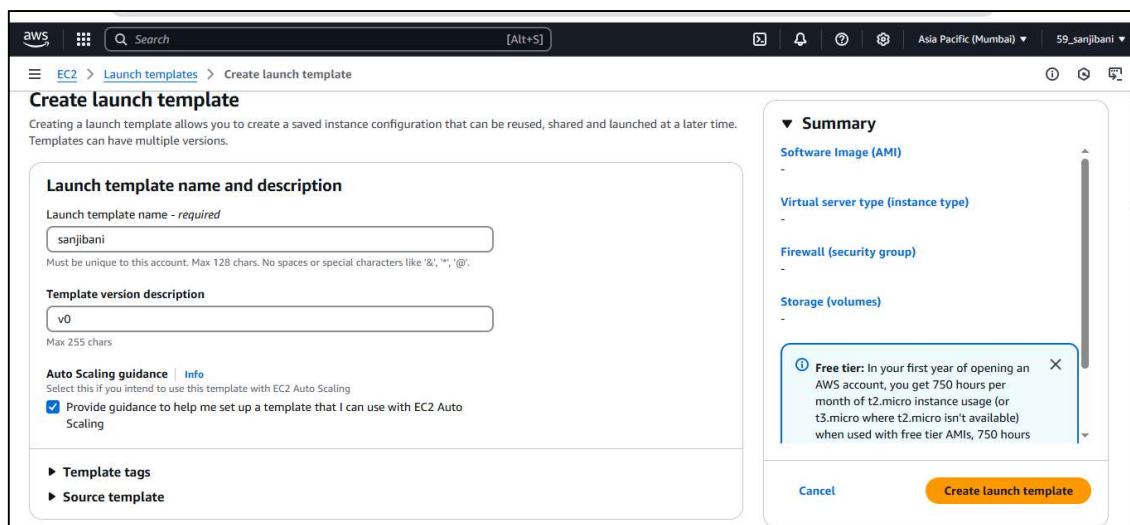
>> And the **Security Group** is Successfully created

The screenshot shows the 'Details' page for the security group 'sg-0dcef9817e2138999 - sanjibani'. A green notification banner at the top states: 'Security group (sg-0dcef9817e2138999 | sanjibani) was created successfully'. The details section shows the security group name, ID, description, VPC ID, owner, inbound rules count (4), and outbound rules count (1). The 'Inbound rules' tab is selected, showing a list of 4 rules.

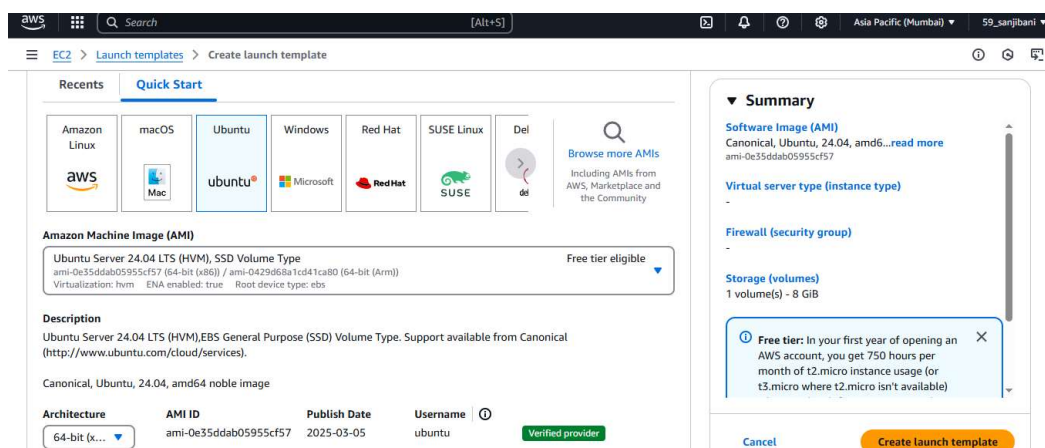
STEP 3 :- Then go to **Launch Template** and click on **Create Launch Template**



STEP 4:- Give your **Template name** and **version description** and select **Auto Scaling guidance**



>> Select Ubuntu in Quickstart



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>> Select **t2.micro** as **Instance type** and create a **key pair**

Instance type Info | Get advice

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true Free tier eligible

On-Demand Linux base pricing: 0.0124 USD per Hour

On-Demand Windows base pricing: 0.017 USD per Hour

On-Demand RHEL base pricing: 0.0268 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0142 USD per Hour

On-Demand SUSE base pricing: 0.0124 USD per Hour

Additional costs apply for AMIs with pre-installed software

Advanced

All generations

Compare instance types

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

Don't include in launch template

Create new key pair

Summary

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more

ami-0e35ddab05955cf57

Virtual server type (instance type)

t2.micro

Firewall (security group)

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available)

Cancel Create launch template

Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

ss11

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA

RSA encrypted private and public key pair

☐ ED25519

ED25519 encrypted private and public key pair

Private key file format

☒ .pem

For use with OpenSSH

☐ .ppk

For use with PuTTY

When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance.

Cancel Create key pair

>> Select the **Security group** you created

Key pair Info

Key pair name

ss11

Create new key pair

Network settings Info

Subnet

Don't include in launch template

Create new subnet

When you specify a subnet, a network interface is automatically added to your template.

Firewall (security groups)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Select existing security group

☐ Create security group

Security groups

Select security groups

sanjibani sg-0dcf9817e2138999

VPC: vpc-064e179da3f6ea9d8

Compare security group rules

Advanced network configuration

Summary

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more

ami-0e35ddab05955cf57

Virtual server type (instance type)

t2.micro

Firewall (security group)

sanjibani

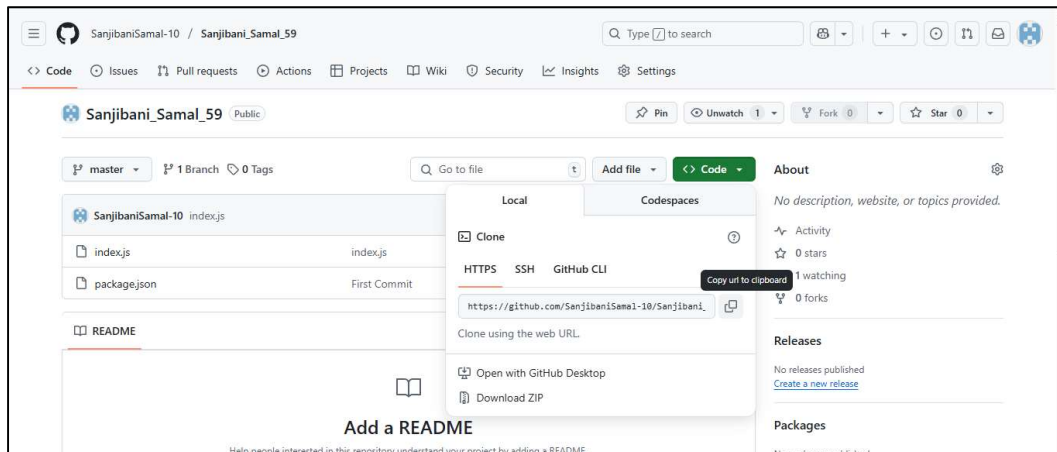
Storage (volumes)

1 volume(s) - 8 GiB

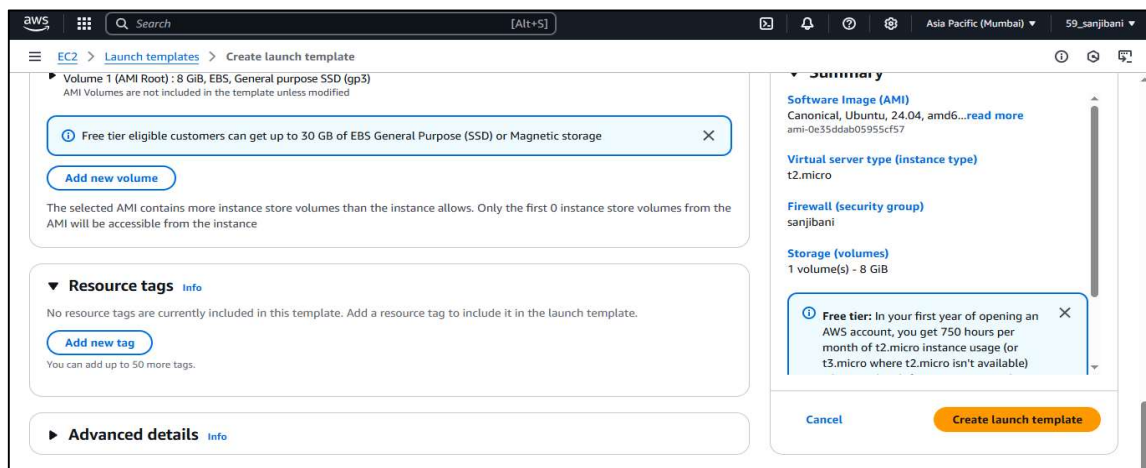
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available)

Cancel Create launch template

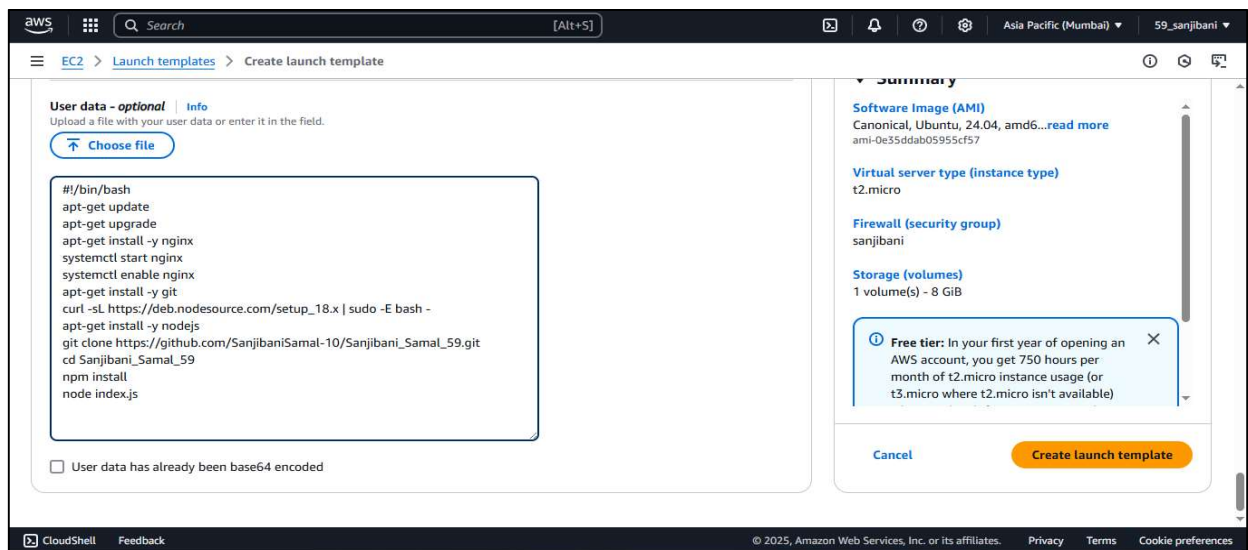
>> Then go to your **Github Repository** and copy the address .



>> then again move to the template createion and click on **Advanced Details**



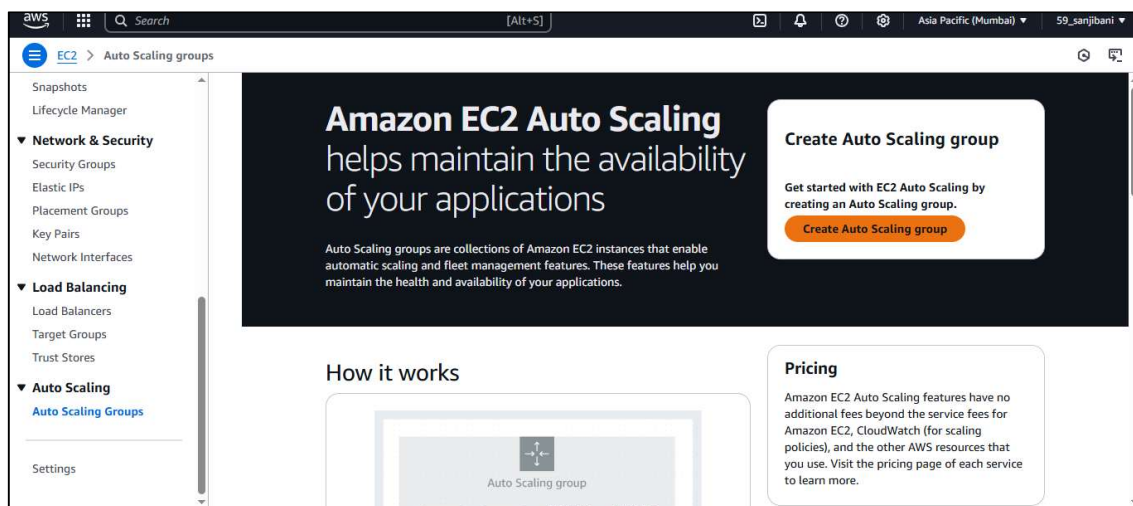
>> write the command line as written below and add your Repository link in **Git Clone**
<Repository Link>



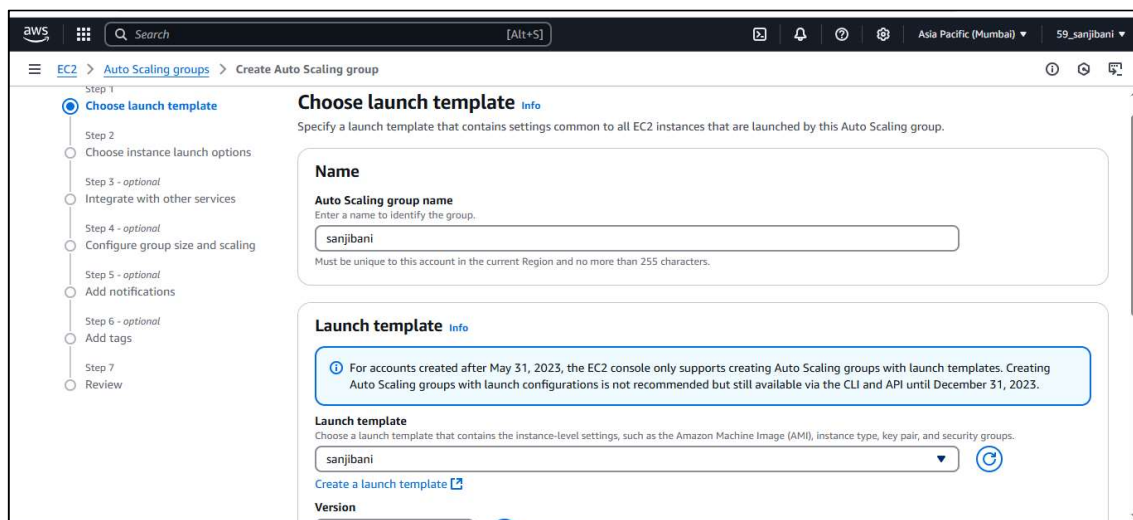
>> Your Template is created successfully.



STEP 5:- Click on **Auto Scaling Groups** and then click on **Create Auto Scaling groups**.



>> Give a **name** and select your desired **Template** and click on Next



>> Give the version as **Latest (1)** and click on Next

aws Search [Alt+S] Asia Pacific (Mumbai) 59_sanjibani

EC2 > Auto Scaling groups > Create Auto Scaling group

Create a launch template

Version: Latest (1) [Create a launch template version](#)

Description: v0

AMI ID: ami-0e35ddab05955cf57

Key pair name: ss11

Launch template: sanjibani [lt-025b9c7a6d0b22c8f](#)

Security groups: -

Security group IDs: sg-0dcef9817e2138999 [sg-0dcef9817e2138999](#)

Instance type: t2.micro

Request Spot Instances: No

Additional details

Storage (volumes): -

Date created: Mon Apr 07 2025 19:49:02 GMT+0530 (India Standard Time)

Cancel Next

>> Select the **Manual add Instance type**

aws Search [Alt+S] Asia Pacific (Mumbai) 59_sanjibani

EC2 > Auto Scaling groups > Create Auto Scaling group

You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.

☐ Specify instance attributes

☒ Manually add instance types

Choose the instance types that best suit the needs of your application.

Primary instance type: t2.micro (1vCPU, 1 GiB Memory)

Additional instance types

Choose the types of instances you want recommended based on the primary instance type

☒ Family and generation flexible ☐ Size flexible

2. t2.small (1vCPU, 2 GiB Memory)

Add instance type

>>In Network region Select the Three Available Zones as servers and click on next

aws Search [Alt+S] Asia Pacific (Mumbai) 59_sanjibani

EC2 > Auto Scaling groups > Create Auto Scaling group

Network

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-064e179da3f6ea9d8 (Default)

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

ap-south-1b | subnet-05008850503b1e74b

ap-south-1a | subnet-0210adc86d82fd1d6

ap-south-1c | subnet-0ff158b56405f6188

Create a subnet

>> Select **Attach a new load balancer** and then select **Network load balancer** and then select **Internet Facing**

The screenshot shows the 'Create Auto Scaling group' wizard in the AWS Management Console. The 'Integrate with other services' step is selected. The 'Load balancing' section has three options: 'No load balancer', 'Attach to an existing load balancer', and 'Attach to a new load balancer'. The 'Attach to a new load balancer' option is selected. Below it, the 'Attach to a new load balancer' section shows 'Load balancer type' with 'Network Load Balancer' selected. The 'Network mapping' section shows 'VPC' as 'vpc-064e179da3f6ea9d8' and 'Availability Zones and subnets' with three subnets selected: 'ap-south-1b', 'ap-south-1a', and 'ap-south-1c'.

>> click on Create Target Group and select a **target group name** and then click on next

The screenshot shows the 'Listeners and routing' step of the 'Create Auto Scaling group' wizard. The 'Protocol' is 'TCP' and the 'Port' is '80'. The 'Default routing (forward to)' section has a dropdown menu with 'Create a target group' selected. The 'New target group name' field is 'sanjibani-1'. The 'Tags - optional' section has an 'Add tag' button. The 'VPC Lattice integration options' section has two options: 'No VPC Lattice service' and 'Attach to VPC Lattice service'. The 'No VPC Lattice service' option is selected.

Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks
[Always enabled](#)

Additional health check types - optional | [Info](#)

☐ Turn on Elastic Load Balancing health checks **Recommended**
 Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

☐ Turn on VPC Lattice health checks
 VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

☐ Turn on Amazon EBS health checks
 EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.

Health check grace period | [Info](#)
 This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

300 seconds

Cancel Skip to review Previous **Next**

>> Now select your **Desired capacity** , **Min desired** and **max desired capacity** and then select **Target Tracking Scaling capacity**.

Step 6 - optional
 Add tags
 Step 7
 Review

Desired capacity
 Specify your group size.
 2

Scaling | [Info](#)
 You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits
 Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity
 2
 Equal or less than desired capacity

Max desired capacity
 3
 Equal or greater than desired capacity

Automatic scaling - optional
 Choose whether to use a target tracking policy | [Info](#)
 You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☐ No scaling policies
 Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ **Target tracking scaling policy**
 Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name

>> Give the **Target Value** and give **Instance Runup** and then click on **Next**

Scaling policy name
 Target Tracking Policy

Metric type | [Info](#)
 Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.
 Average CPU utilization

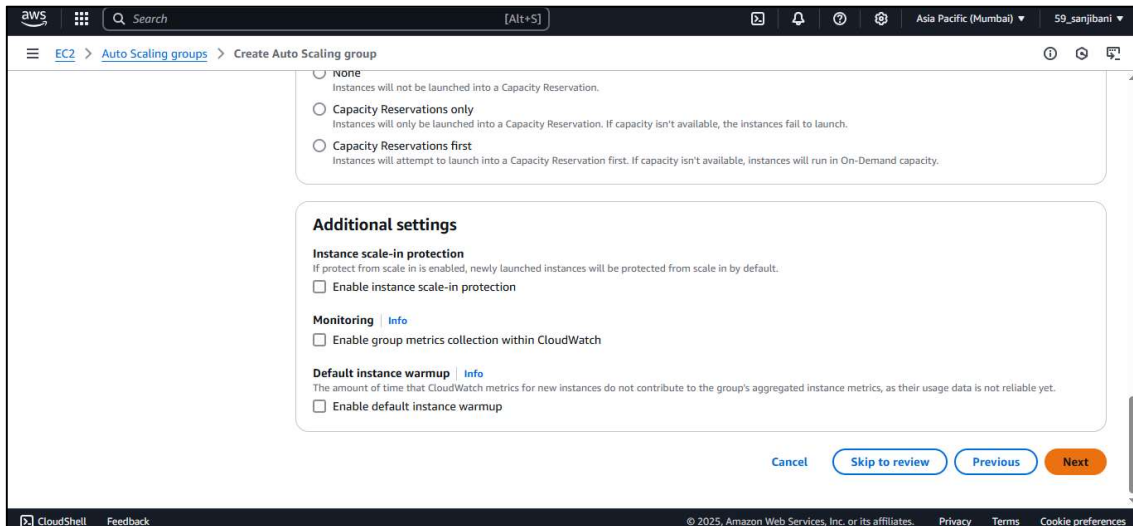
Target value
 32

Instance warmup | [Info](#)
 300 seconds

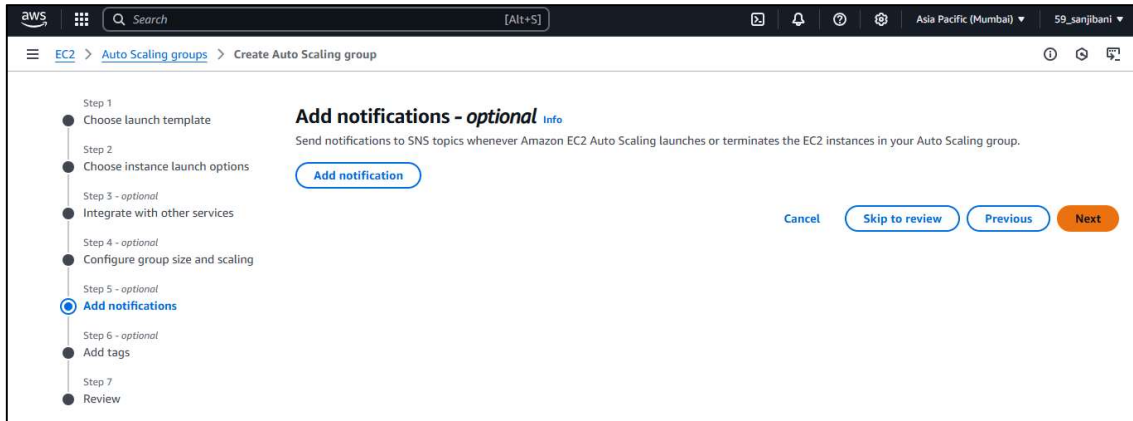
☐ Disable scale in to create only a scale-out policy

Instance maintenance policy | [Info](#)
 Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.

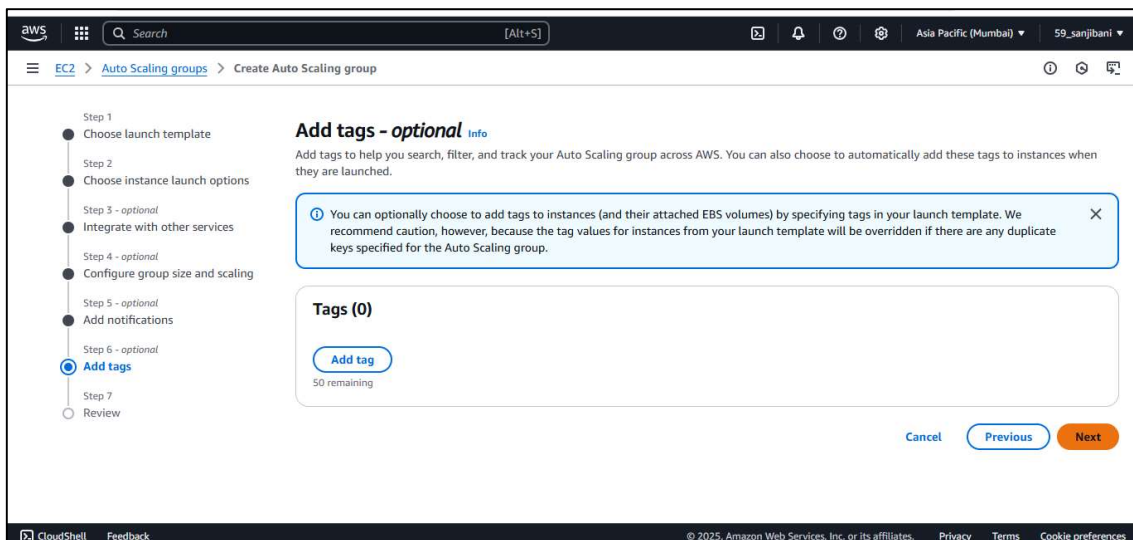
Choose a replacement behavior depending on your availability requirements



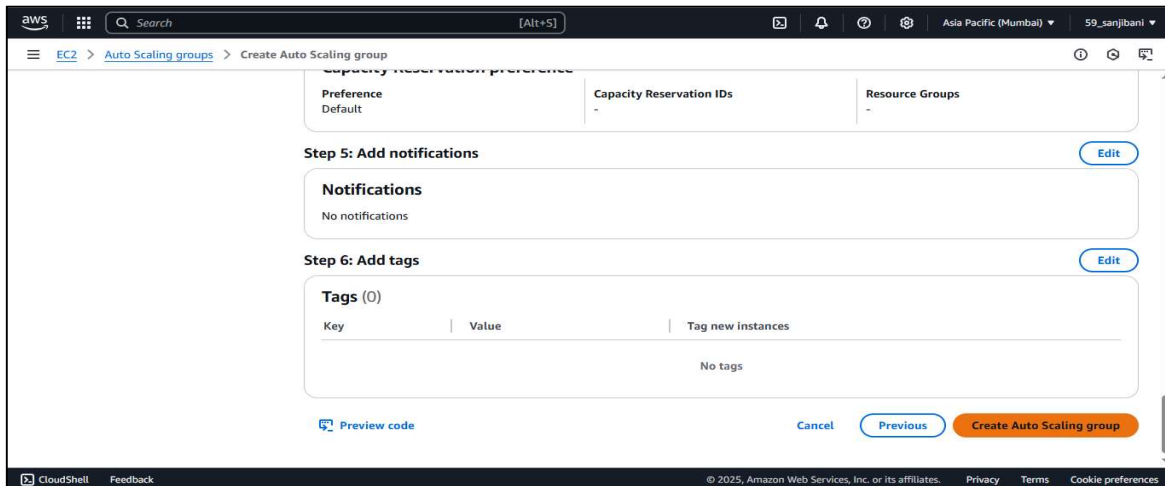
>>click on **NEXT**



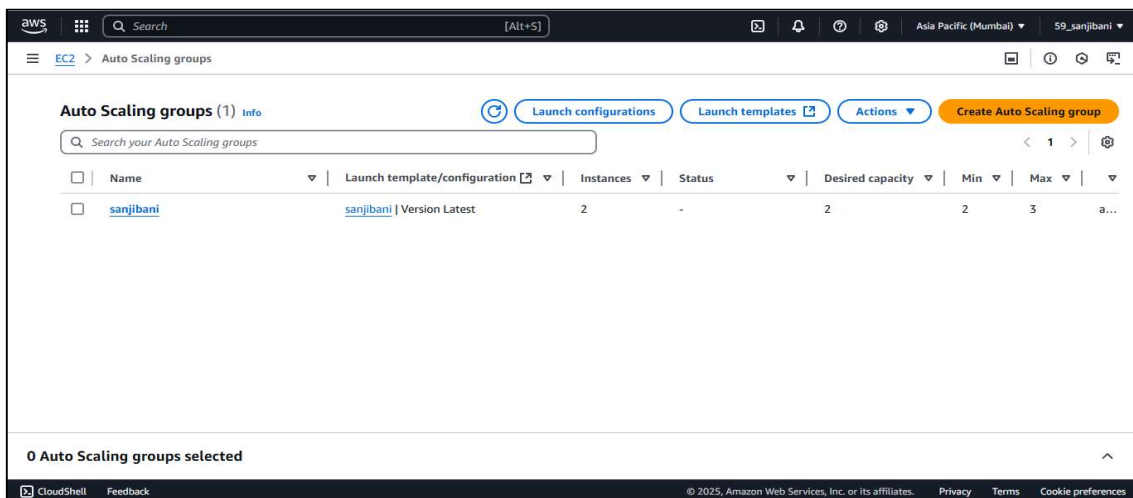
>>click on **NEXT**



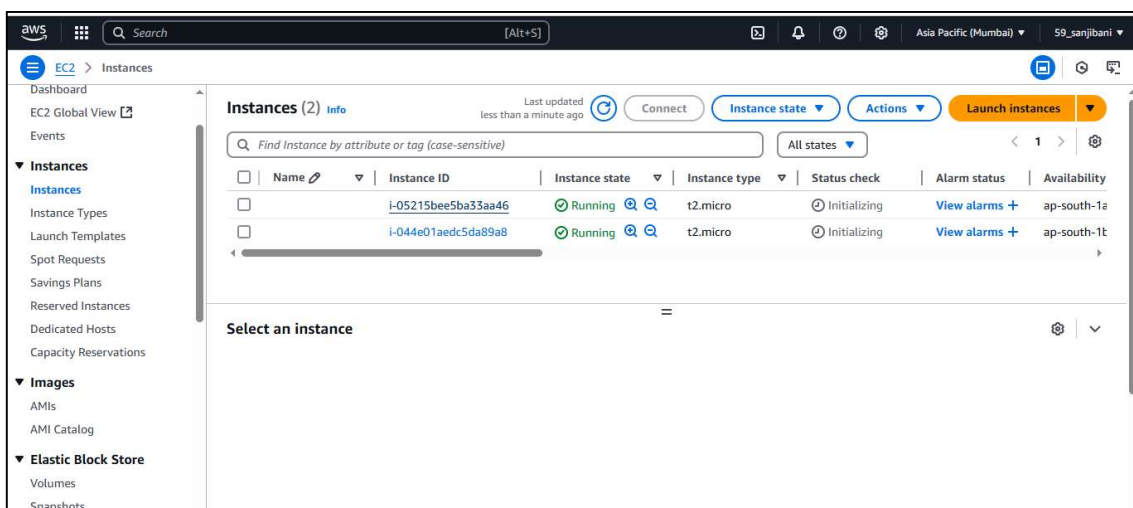
>>click on **Create Auto Scaling Group**



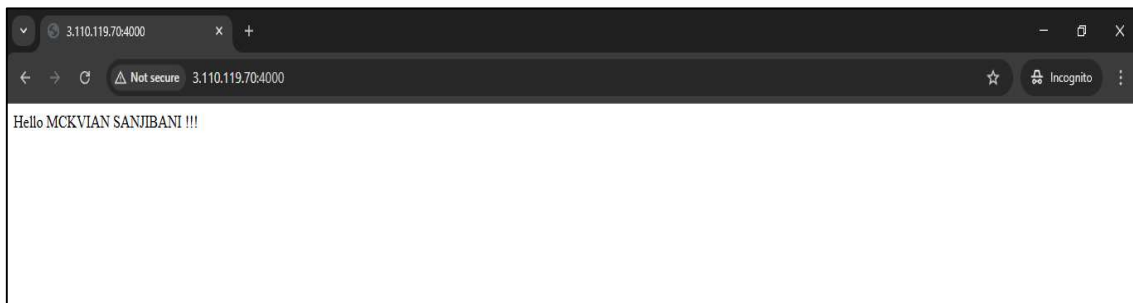
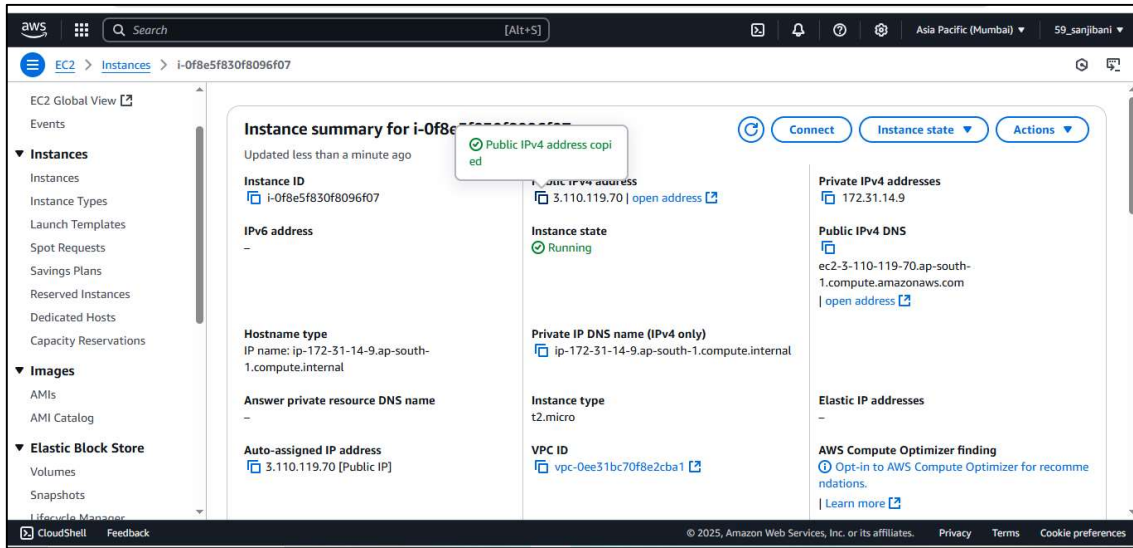
>> And your Auto Scaling Group is Created Successfully



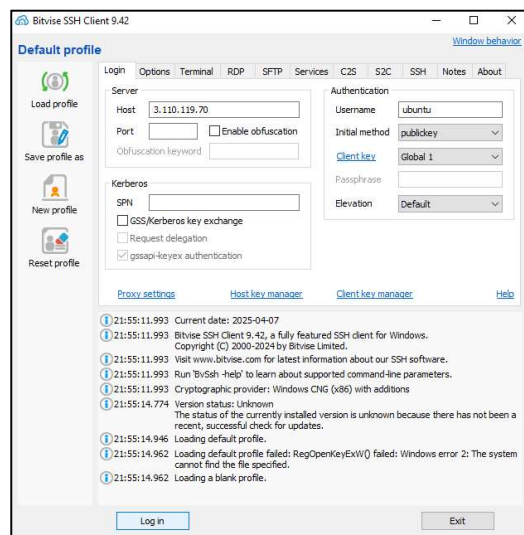
STEP 5:- As you can see the **Instances are automatically created** if you will delete one instance then the 3rd one will automatically generated as **3** servers are given



STEP 6:- Select any one **Instance** and Copy its **IPv4 address** and paste it in **Incognito website** with port number(**IP Address : 4000**)



STEP 7:- Paste the address in **Bitwise Server** and give the required field of **username** , **Initial method** and **Client key** then click on **LOG IN** and go to the **New Terminal**



>> Then write the following command in terminal to open a shell file

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>> Then write the following code in the shell file

aws

[Alt+S]

Search

Asia Pacific (Mumbai)

59_sanjibani

```
GNU nano 7.2 infil.sh *  
#!/bin/bash  
while true  
do  
echo "looping"  
done
```

Help

Write Out

Where Is

Cut

Execute

Location

Undo

Set Mark

To Bracket

Exit

Read File

Replace

Paste

Justify

Go To Line

Redo

Copy

Where Was

i-04361640d3d0da927

PublicIPs: 13.127.248.240 PrivateIPs: 172.31.39.192

CloudShell

Feedback

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Privacy

Terms

Cookie preferences

>> Compile and run the File Using the following command

```
aws
```

☰

Q Search

[Alt+S]

🔍 🔔 ⚙️ 🌐

Asia Pacific (Mumbai) 59_sanjibani

Expanded Security Maintenance for Applications is not enabled.
49 updates can be applied immediately.
23 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-39-192:~\$ nano infil.sh
ubuntu@ip-172-31-39-192:~\$ chmod +x infil.sh
ubuntu@ip-172-31-39-192:~\$./infil.sh

i-04361640d3d0da927

PublicIPs: 13.127.248.240 PrivateIPs: 172.31.39.192

The screenshot shows an AWS CloudShell terminal window. The top bar includes the AWS logo, navigation icons, a search bar, and the text "[Alt+S]". The right side of the bar shows the region "Asia Pacific (Mumbai)" and the user "59_sanjibani". The terminal output consists of 20 lines of "ping" commands, followed by the IP address "i-04361640d3d0da927". The bottom status bar displays the public IP "PublicIP: 13.127.248.240" and the private IP "PrivateIP: 172.31.39.182".

```

aws
[Alt+S]
Asia Pacific (Mumbai)
59_sanjibani
ping
ping
ping
ping
ping
ping
ping
ping
ping
ping
ping
ping
ping
ping
ping
ping
ping
ping
ping
ping
ping
i-04361640d3d0da927
PublicIP: 13.127.248.240 PrivateIP: 172.31.39.182

```

>> Then click on both the instances and you can see the graph of both processes **CPU UTILISATION**

